

■ 6-Month Data Engineering Roadmap

Data Engineering is one of the fastest-growing careers in technology. With AI and modern data solutions driving businesses, the role of Data Engineers is critical. They design scalable pipelines, manage distributed data, and ensure that information flows reliably across systems. Without them, analytics, machine learning, and AI cannot function effectively. Demand is booming: LinkedIn lists over 180,000+ Data Engineering jobs globally. Salaries are highly competitive, ranging from solid entry-level offers to over \$450,000 in the US and ■1–2 crores in India for senior roles. This roadmap outlines a 6-month structured plan to transition into Data Engineering with skills, projects, and job readiness.

■ Phase 1: Months 1–4 → Build Skills

1. Python (Month 1)

Why: Foundation of data engineering (ETL, Spark, Airflow, cloud SDKs).

Resources:

- ➡■ Automate the Boring Stuff — Beginner-friendly eBook with real-world Python examples.
- ➡■ Python Data Engineer Track – DataCamp — Covers Python, databases, APIs, Git, and ETL/ELT basics.

2. SQL & DBMS (Month 2)

Why: SQL is the superpower for querying, modeling, and analytics. DBMS concepts form the foundation.

Resources:

- ➡■ DBMS Crash Course – StudyTonight — Learn fundamentals like normalization and transactions.
- ➡■ SQL Course – DataCamp — From basics to advanced queries with hands-on practice.
- ➡■ SQL Certification – DataCamp — Practical certification with real-world case studies.

3. Cloud (Month 3)

Why: Cloud is non-negotiable in modern data engineering. Start with AWS for widest adoption and jobs.

Focus Services: S3, EC2, ECS, VPCs, RDS, DynamoDB, Redshift, EMR, Glue, Lambda, Kinesis.

Resources:

- ➡■ AWS Solutions Architect Course – Udemy — Complete AWS foundations course by Stefan Marek.
- ➡■ Google Cloud Skills Boost — Hands-on labs for GCP.
- ➡■ Azure DP-700 Full Course (YouTube) — Covers Microsoft's data-focused certification.

4. Modern Data Stack (Month 4)

Why: Handle terabytes of data efficiently with distributed & orchestration tools.

Resources:

- ➡■ Apache Spark Course – Udemy — Learn Spark with hands-on projects.
- ➡■ Apache Kafka Documentation — Official guide for real-time streaming.
- ➡■ Apache Airflow Documentation — Learn orchestration and scheduling.
- ➡■ Flink Basics – Confluent — For ultra-low latency stream processing (optional).

■ Phase 2: Month 5 → Projects & Portfolio

Why: Projects prove your skills more than certificates. Build one batch and one streaming pipeline.

Resources:

- ➡■ Data.gov — Public datasets for batch projects.
- ➡■ AWS Open Data — Free datasets on AWS.
- ➡■ Azure Open Datasets — Curated datasets from Microsoft.
- ➡■ GCP BigQuery Public Data — Free-to-query datasets on Google Cloud.
- ➡■ Awesome Public Realtime Datasets — Streaming datasets for practice.
- ➡■ Batch Project Guide — Step-by-step project with Spark, Airflow, DB.
- ➡■ Realtime Project Guide — Beginner-friendly streaming project with Kafka/Flink.

■ Phase 3: Month 6 → Job Preparation & Hunt

Why: Portfolio + visibility + interview prep = job readiness.

- **Portfolio:** GitHub with 2–3 projects, architecture diagrams, and clear READMEs.
- **Branding:** Blog posts (Medium, LinkedIn) showcasing your projects.
- ➡■ LeetCode — Practice SQL & DSA.
- Focus on Data Modeling (schemas, normalization, fact/dimension tables).
- Practice System Design (scalable pipelines, orchestration).

■ Study Calendar (Checklist)

Month	Focus Area	Checklist
1	Python	■ Learn Python basics ■ Automate the Boring Stuff ■ DataCamp Python Track
2	SQL & DBMS	■ Study DBMS ■ Learn SQL ■ SQL Certification
3	Cloud	■ Cover AWS services ■ Explore GCP/Azure basics ■ Do 1 small cloud lab
4	Modern Data Stack	■ Learn Spark ■ Kafka basics ■ Airflow orchestration ■ Optional Flink
5	Projects	■ Batch project complete ■ Streaming project complete ■ Deploy on cloud
6	Job Prep	■ GitHub portfolio ■ Blog posts ■ Resume update ■ Mock interviews

■ **Outcome:** If followed consistently, this roadmap helps you transition into a Data Engineering role in ~6 months, with both technical depth and portfolio strength to crack interviews.